Statement of Purpose [Agriculture Project]

Agriculture forms the backbone of many economies, particularly in developing countries, where a significant portion of the population relies on farming for their livelihood. However, crop diseases pose a significant threat to agricultural productivity, leading to substantial financial losses and food insecurity. Early detection and accurate classification of crop diseases are essential to minimizing their impact and ensuring sustainable farming practices. Traditional methods of disease diagnosis, such as manual inspection by experts, are time-consuming, subjective, and often limited by the availability of trained personnel.

With the rapid advancement in computer vision and artificial intelligence, automated disease detection using image-based analysis has emerged as an effective solution. Deep neural networks (DNNs), a subset of machine learning techniques, have shown remarkable success in various image classification tasks, including medical imaging, object recognition, and agricultural monitoring. However, the performance of these models is highly dependent on the quality of input images. Real-world agricultural images are often affected by factors such as varying lighting conditions, background noise, blur, and low contrast, which can obscure disease patterns and reduce classification accuracy.

To address these challenges, image enhancement techniques are employed to preprocess and refine the captured images. Enhancing image contrast, removing noise, and emphasizing key features can significantly improve the ability of deep learning models to learn disease-specific patterns. By integrating image enhancement with deep neural networks, it becomes possible to create a robust and scalable system that can detect and classify diseases at an early stage, providing timely support to farmers and agricultural stakeholders.

This research aims to develop a comprehensive framework that combines image enhancement and deep learning algorithms for early disease detection and classification. The proposed system leverages preprocessing techniques to improve image quality and uses deep neural networks to accurately classify diseases from enhanced images. The results are expected to contribute to the development of practical tools that can be deployed in real-world agricultural environments, helping farmers to take preventive actions and improve crop management.

Objective and Scope of the Research Work

1. **Objective 1:** Dataset creation by collecting real-time field images using devices such as mobile phones, digital cameras, and drones, along with controlled laboratory images.
2. **Objective 2:** Applying different deep learning and computer vision algorithms to the collected crop images**.**
3. **Objective 3:** To develop a robust and efficient model for crop disease classification and identification using artificial intelligence techniques**.**

Scope of the Research Work:

1. Develop scalable solutions for real-time crop disease detection on mobile phones, edge devices, and drones.
2. To integrate explainable AI methods that provide transparent and interpretable predictions for farmers and agronomists.
3. Agricultural expert feedback and field verification are used to retrain and fine-tune the model, improving disease prediction reliability.

For agriculture applications, especially plant disease detection/classification, crop monitoring, weed detection, and leaf-image analysis, you can apply the deep-learning/CNN techniques as follows.

1. Traditional CNN Architectures

- **LeNet**
- **AlexNet**
- **VGG16 / VGG19**
- **GoogLeNet / Inception**
- **InceptionV3**
- **Inception-ResNet**
- **ResNet18 / ResNet34 / ResNet50 / ResNet101**
- **DenseNet121 / DenseNet169 / DenseNet201**
- **SqueezeNet**
- **Xception**
- **MobileNet / MobileNetV2 / MobileNetV3**
- **ShuffleNet**

2. Advanced CNN Models

- **EfficientNet / EfficientNetV2**
- **ConvNeXt**
- **RegNet**
- **ResNeXt**
- **SE-Net (SENet)**
- **NASNet**
- **GhostNet**

3. Object Detection Algorithms

Useful when the objective is to locate diseases, fruits, weeds, or plant parts in an image:

- **R-CNN**
- **Fast R-CNN**
- **Faster R-CNN**
- **Mask R-CNN**
- **SSD (Single Shot Detector)**
- **YOLOv3**
- **YOLOv4**
- **YOLOv5**
- **YOLOv7**
- **YOLOv8**
- **YOLOv9**
- **YOLOv10**
- **YOLOv11**
- **RetinaNet**

4. Image Segmentation Techniques

Useful for identifying the exact diseased region or leaf area:

- **U-Net**
- **U-Net++**
- **SegNet**
- **DeepLabV3**
- **DeepLabV3+**
- **FCN**
- **PSPNet**
- **Mask R-CNN**

5. Transfer Learning Approaches

Very common in agricultural image classification when the available dataset is limited:

- **VGG + Transfer Learning**
- **ResNet + Transfer Learning**
- **Inception + Transfer Learning**
- **DenseNet + Transfer Learning**
- **MobileNet + Transfer Learning**
- **EfficientNet + Transfer Learning**
- **Xception + Transfer Learning**

6. CNN-Based Hybrid/Attention Models

These can improve the model's ability to focus on important disease features:

- **CNN + Attention Mechanism**
- **CNN + SE Block**
- **CNN + CBAM**
- **CNN + Transformer**
- **CNN + LSTM**
- **CNN + BiLSTM**
- **Multi-scale CNN**
- **Ensemble CNN**

7. Transformer-Based Models

Increasingly used in agricultural computer vision:

- **Vision Transformer (ViT)**
- **Swin Transformer**
- **DeiT**
- **MobileViT**
- **ConvNeXt + Transformer hybrids**

8. Generative and Augmentation-Based Techniques

Useful for increasing training data and handling limited agricultural datasets:

- **GAN (Generative Adversarial Network)**
- **DCGAN**
- **Conditional GAN (cGAN)**

- **StyleGAN**
- **Diffusion-based image generation**
- **Mixup**
- **CutMix**
- **Random Erasing**

[In Addition you have any idea add on it]